

Analysis of portfolio diversification benefits on selected companies monthly closing returns

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Executive Summary

This report analyzes the risk-return profile of a 10-stock portfolio (AAPL, BAC, COST, HD, LLY, MA, META, MSFT, PG, UNH) compared to the SPY over the five-year period ending December 31, 2025 (monthly data from December 2020).

Findings show that equally weighted portfolio delivers a strong annualized mean return of 17.87% with lower volatility of 14.92% than most individual stocks, demonstrating clear diversification benefits. Portfolio's Sharpe ratio of 0.87 exceeds SPY's 0.76, indicating superior risk-adjusted performance. Beta of the portfolio 0.87 is below 1.00, indicating reduced market sensitivity due to defensive holdings. Cost of equity estimates using CAPM range from 5.4% in pharmaceutical to 10.3% in large bank.

Data for these analyses were source from Yahoo Finance adjusted close prices and Professor Francisca Rabelo provided risk-free rates APR which was converted to monthly EAR.

Data and Methodology

Monthly adjusted closing prices were obtained for December 2020 - December 2025. Monthly returns calculated as:

$$R_t = (P_t / P_{(t-1)}) - 1$$

Excess returns = monthly return – monthly risk-free rate (Rf monthly = R-F APR / 12). Equally weighted portfolio return = simple average of the 10 stocks' monthly returns (SPY excluded).

Statistics computed over 60 monthly observations include Mean and standard deviation (sample STDEV.S); Annualized: mean $\times$ 12; stdev $\times \sqrt{12}$; Sharpe ratio = (annual excess mean) / (annual excess stdev); Correlation and beta via CORREL and SLOPE on excess returns vs. SPY excess.

Key Results and Observations

Risk-Return Plot

Figure 1

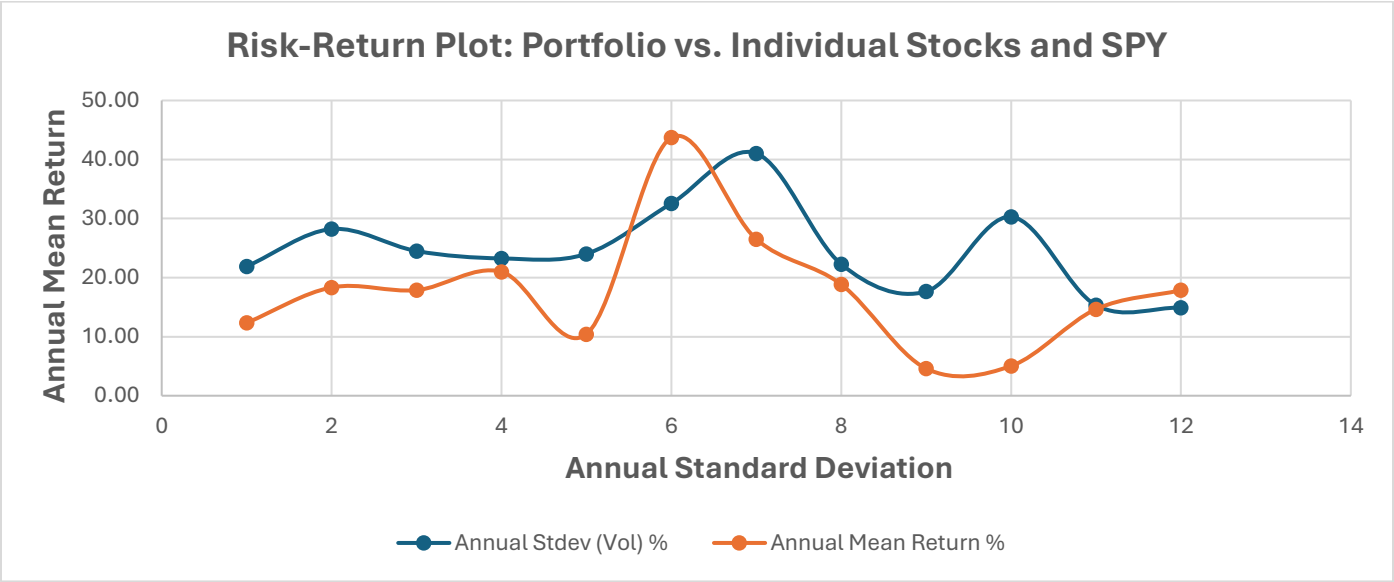


Figure 1 above reveals a risk-return tradeoff, it was observed that Individual stocks cluster with higher volatility between 20 to 41% with varying returns ranging from 10 to 44% annually. Exhibit 4 (statistics) shows that META stands out with the highest volatility of 41.04% and strong return of 26.52%, reflecting tech growth

exposure. Eli Lilly (LLY) shows exceptional return of 43.72% at moderate volatility of 32.56%, driven by pharmaceutical innovation. The equally weighted portfolio (Orange points) positioned at lower volatility of 15% than most stocks while maintaining high return of about 18%, closer to SPY but with better efficiency. The plot above shows that diversification reduces unsystematic risk without sacrificing much overall return.

Annualized Volatility

Figure 2

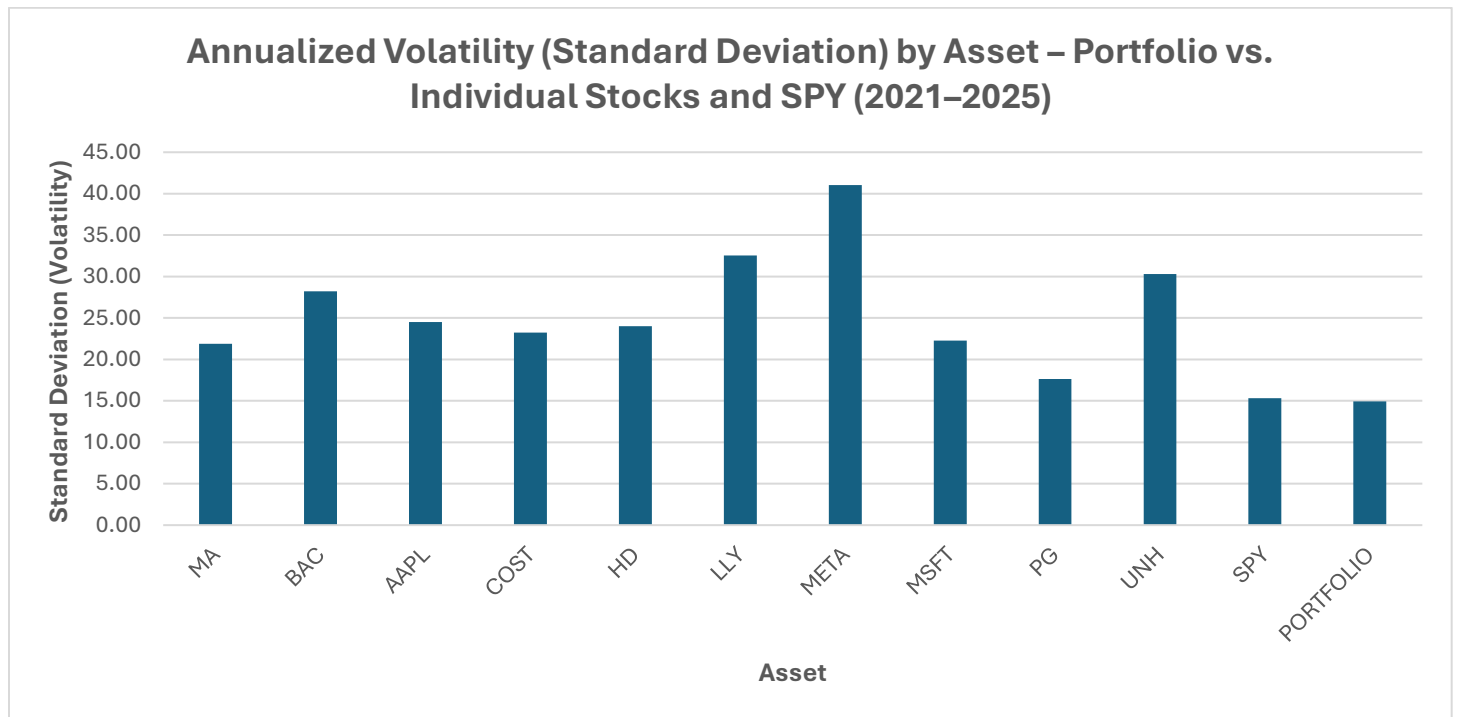
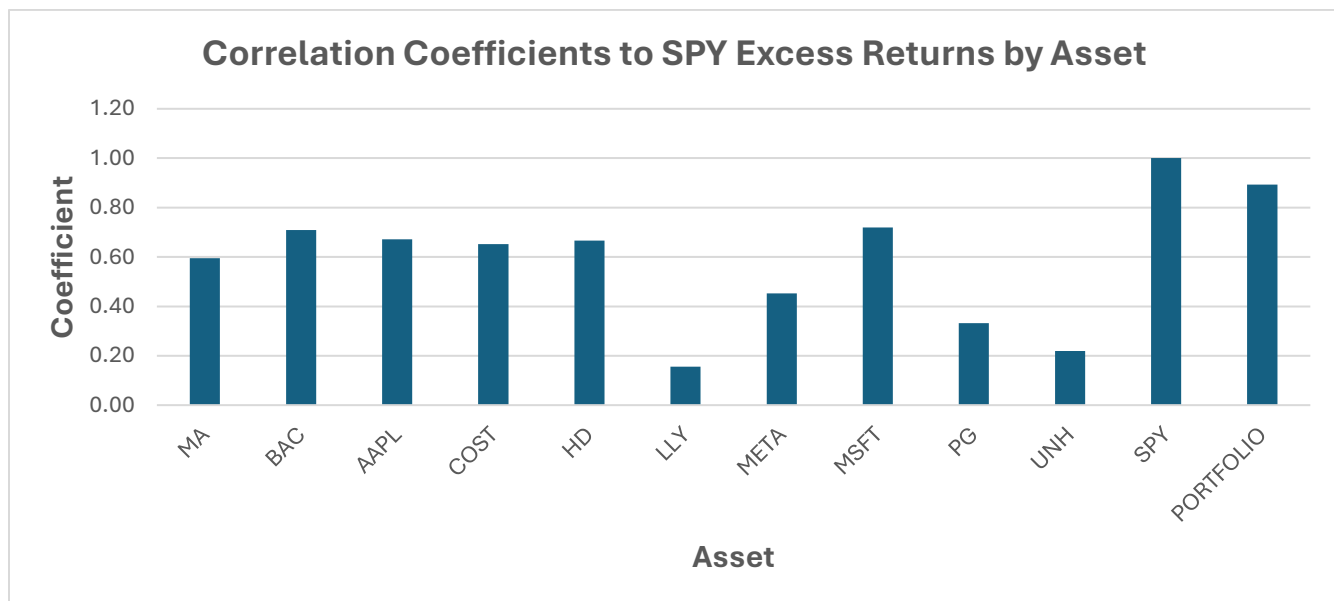


Figure 2 above shows that META exhibits the highest volatility of 41% consistent with tech sector swing. Procter and Gamble (PG) shows a volatility of about 17.65% while Costco (COST) shows 23.26%, indicating a lower volatility. Total Portfolio volatility of 14.92% is the lowest overall which is tied to SPY volatility of 15.33%, confirming diversification reduces total risk substantially.

Correlation to SPY

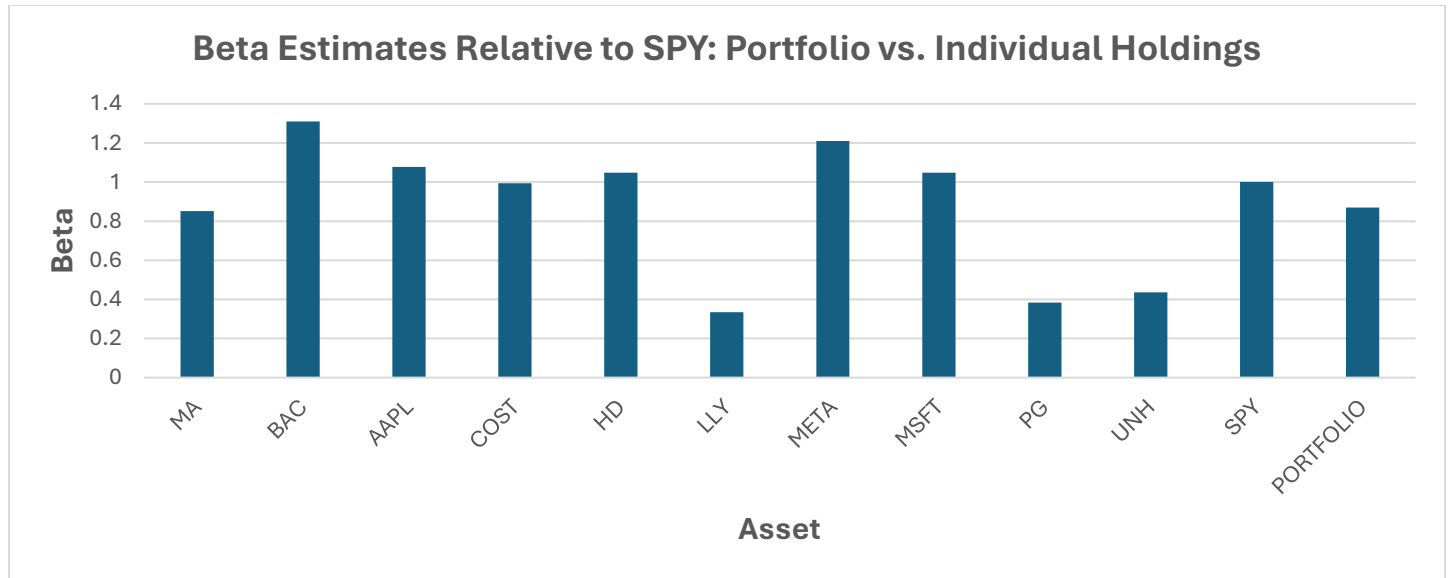
Figure 3



Most assets in Figure 3 show moderate to high correlation (0.6–0.9), Eli Lilly (LLY) has very low correlation of 0.16, indicating independence from broader market movements likely due to pharma growth. Portfolio correlation of 0.89 is high but below 1.00, reflecting blended exposure.

Beta to SPY

Figure 4



It was observed in figure 4 above that Bank of America (BAC) with a beta of 1.31, and META with a beta of 1.21, show highest betas. Eli Lilly (LLY) with 0.33 and Procter and Gamble (PG) with 0.38 are the lowest.

Portfolio beta of $0.87 < 1.00$ indicates lower systematic risk than the market, which is beneficial for a risk averse client.

Cost of Capital Forecasts (CAPM)

Using forecast $R_f = 3.75\%$ and $MRP = 5\%$, cost of equity = $R_f + \beta \times MRP$.

Betas selected/averaged from similar portfolio stocks:

Industry	(TIC) Company selected	Beta	Average beta used	Cost of capital
National big-box retailer	COST (0.99); HD (1.05)	$0.99 + 1.05$	$(0.99+1.05)/2 = 1.02$	8.85%
Pharmaceutical	LLY (0.33)	0.33		5.42%
Large Bank	BAC	1.31		10.30%
Technology	AAPL(1.08); MSFT (1.05); META (1.21)	$1.08+1.05+1.21$	$(1.08+1.05+1.21)/2 = 1.11$	9.30%

****The Excel workbook contains all supporting calculations and charts for verification. ****